SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE OUTCOME COVERED

COURSE CODE: CSA15

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) Question Count: 5

Duration : 60 Minutes **Code**

of Conduct

Total Marks: 50

I __**Shaik Mohammad akhim(192525132)**_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: _____**Shaik Mohammad akhim**_____

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces. 3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

• Library pane showing multiple VMs = multiple instances/regions.

• VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.

• Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.

• VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

ANSWERS:

1. Compare Type-1 and Type-2 Hypervisors for a Cloud Datacenter

Type-1 Hypervisor (Bare-Metal):

- Runs directly on physical hardware.
- Provides high performance and efficiency.
- Offers better security and isolation.
- Suitable for large cloud datacenters.
- Example: VMware ESXi, Microsoft Hyper-V.

Type-2 Hypervisor (Hosted):

- Runs on top of an operating system.
- Performance is slightly lower due to the host OS.
- Depends on the host operating system for security.
- Suitable for personal use, testing and development.
- Example: VMware Workstation Pro, Oracle VirtualBox.

Feature	**Type-1**	**Type-2**
Performance	High	Moderate
Security	Strong	Depends on host OS
Management	Datacenter-level	Desktop-level
Hardware access	Direct	Through host OS
Usage	Cloud servers	Personal systems

Conclusion: Type-1 hypervisors are the better choice for mission-critical cloud applications because they provide better performance, security, scalability and reliability.

2. Workload Isolation in a Cloud Provider

A cloud provider may host finance, healthcare and student portal applications on the same physical server.

How virtualization supports isolation:

1. Each application runs inside a separate virtual machine.
2. Each VM gets its own operating system and virtual resources.
3. Hypervisors control CPU, memory and storage access.
4. Virtual networks separate communication between applications.
5. Access permissions prevent unauthorized users from accessing other VMs.

Two major attack surfaces:

- Hypervisor attacks: Attackers may exploit hypervisor vulnerabilities to access multiple virtual machines.
- Virtual network attacks: Attackers may use network vulnerabilities to access or intercept data between VMs.

Security measures:

- Use strong authentication and access control.
- Regularly update the hypervisor.
- Apply network segmentation and firewalls.
- Encrypt sensitive data.

Conclusion: Virtualization provides workload isolation and improves security by separating different applications and controlling resource access.

3. VM Host Utilization Analysis

Given utilization snapshot:

Resource	Utilization	Analysis
CPU	92%	Very high usage
Memory	88%	High memory consumption
Storage I/O latency	35 ms	Slow storage operations
VM boot time	170 seconds	Slow VM startup

Likely bottlenecks:

1. CPU bottleneck: High CPU utilization can cause slow processing and reduced VM performance.

2. Memory bottleneck: High memory usage may cause swapping and slow application execution.
3. Storage bottleneck: 35 ms I/O latency indicates slow disk operations.
4. Boot-time issue: Slow storage and resource contention may increase VM startup time.

Corrective actions:

- Add more CPU cores or upgrade the processor.
- Increase RAM capacity.
- Use SSD or NVMe storage.
- Distribute VMs across multiple hosts.
- Remove unused virtual machines.
- Monitor resource utilization regularly.

Conclusion: The host is overloaded. Increasing resources, improving storage performance and balancing VM workloads will improve system performance.

4. Software Defined Datacenter (SDDC) Architecture

A Software Defined Datacenter is a datacenter where computing, storage and networking resources are controlled through software.

How SDDC improves agility:

1. Compute Virtualization

- Creates multiple virtual machines on one physical server.
- Resources can be allocated dynamically.
- Makes server management easier.

2. Storage Virtualization

- Combines multiple storage devices into a single storage pool.
- Provides flexible storage allocation.
- Improves storage utilization and backup management.

3. Network Virtualization

- Creates virtual networks using software.
- Supports network segmentation.
- Allows faster network configuration.

Comparison:

Traditional Datacenter	Software Defined Datacenter
Manual management	Software-based management
Hardware dependent	Virtualization based
Slow resource allocation	Fast resource allocation
Less flexible	Highly flexible
Difficult scaling	Easy scaling

Conclusion: SDDC improves agility by automating resource management, reducing manual work and providing faster scalability.

5. Multi-Cloud Federation and Identity Mismatch

Problem:

A multi-cloud federation project connects services from different cloud providers. The project fails because each provider uses different identity formats, authentication methods and access policies.

Reasons for identity mismatch:

1. Different user ID formats.
2. Different authentication protocols.
3. Different access permissions.
4. Incompatible security policies.

Proposed secure identity and privacy design:

1. Single Sign-On (SSO): Allows users to access multiple cloud services using one login.
2. Identity Federation: Connects identity systems of different cloud providers.
3. SAML and OpenID Connect: Use standard protocols to exchange authentication information.
4. Role-Based Access Control (RBAC): Provides permissions based on user roles.
5. Multi-Factor Authentication (MFA): Adds an extra security layer.
6. Data Encryption: Protects sensitive information during storage and transmission.
7. Privacy Protection: Share only the minimum required user information between providers.

Conclusion: Identity federation, standard authentication protocols and proper access control provide secure communication between multiple cloud providers while protecting user privacy.

Analytical Problem Solving Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 – Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Wellstructured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded: